DEPARTAMENTO DE ELETRÓNICA, TELECOMUNICAÇÕES
E INFORMÁTICA

LICENCIATURA EM ENGENHARIA DE COMPUTADORES E INFORMÁTICA

ANO 2025/2026

MÉTODOS PROBABILÍSTICOS PARA ENGENHARIA DE COMPUTADORES E INFORMÁTICA

PRACTICAL GUIDE NO. 3

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Note: In all cases of this guide, adopt the definition of the state transition matrix T in which the matrix element t_{ij} corresponds to the transition probability from state j to state i.

1. Consider that the weather on each day of January is generally classified into one of 3 states - Sunny, Cloudy and Rainy - and that the weather on a given day only depends on the weather on the previous day as specified in the table below. Assume that the 1st of January is a Wednesday.

de

para

day n \ day $n+1$ →	Sunny	Cloudy	Rainy
Sunny	0.7	0.2	0.1
Cloudy	0.2	0.3	0.5
Rainy	0.3	0.3	0.4

100 - 31
45 - k

(a) Define, in Matlab, the corresponding state transition matrix T.

(b) What is the probability that the second and third days of January are both sunny, when the first day is sunny?

(c) What is the probability of being sunny in both days of the first weekend of January, knowing that the first day is rainy?

(d) Assuming that the first day is sunny, determine the expected average number of sunny, cloudy and rainy days throughout January.

(e) Assuming that the first day is rainy, determine the expected average number of sunny, cloudy and rainy days throughout January. Compare these results with those obtained in the previous question. What do you conclude?

(f) Consider a person with chronic rheumatism who has rheumatic pain with probabilities of 10%, 30% and 50% when the days are sunny, cloudy or rainy, respectively. What is the expected number of days that a person will suffer from rheumatic pain in January when the first day is sunny? And when the first day is rainy?

2. (Optional) Consider the following exchange of students between groups: a class is divided into 3 groups (A, B and C) at the beginning of the semester. At the end of each class the following student movements are made:

- 1/3 from group A goes to group B and another 1/3 from group A goes to group C,
- 1/4 from group B goes to group A and 1/5 from group B goes to group C,
- half of group C goes to group B and the other half remains in group C.

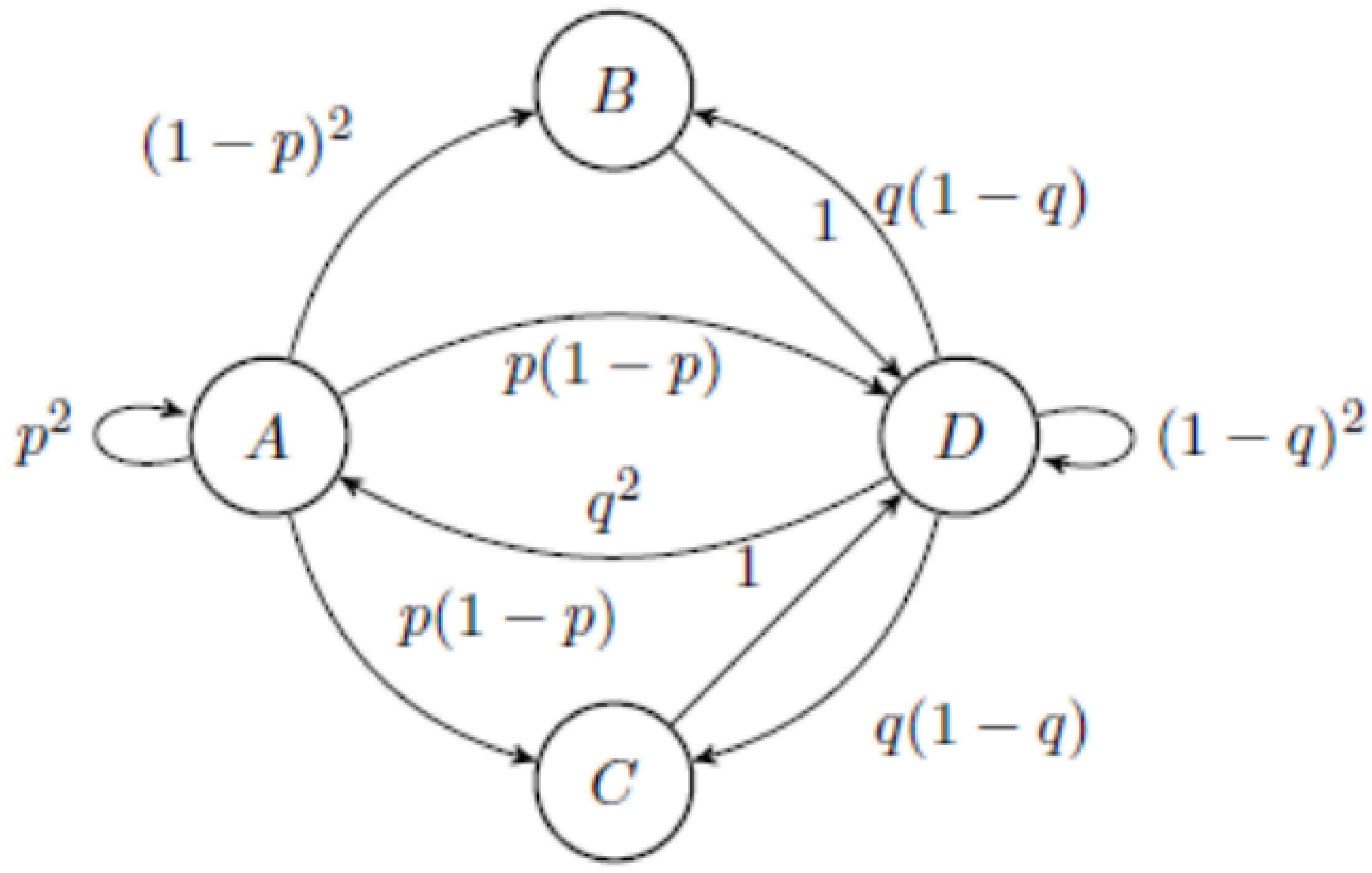
(a) Create in Matlab the state transition matrix T that represents the exchanges between groups. Confirm that T is stochastic.

(b) Create the vector describing the initial state considering that: there are 90 students in total, group A has twice the sum of the other two groups, and groups B and C have the same number of students.

(c) How many students will be in each group at the end of class 30 considering the same initial state as defined previously?

(d) How many students will be in each group at the end of class 30 considering that initially the 90 students were equally distributed among the 3 groups?

3 Consider the following diagram representing a discrete time Markov Chain:



(a) Define, in Matlab, the state transition matrix T assuming $p = 0.4$ and $q = 0.6$.

(b) Assume that the system is initially in state A. What is the probability of the system being in each state after 5 transitions? And 10 transitions? And 100 transitions? And 200 transitions?

(c) Determine the limit probabilities of each state. Compare these values with those obtained in the previous question. What do you conclude?

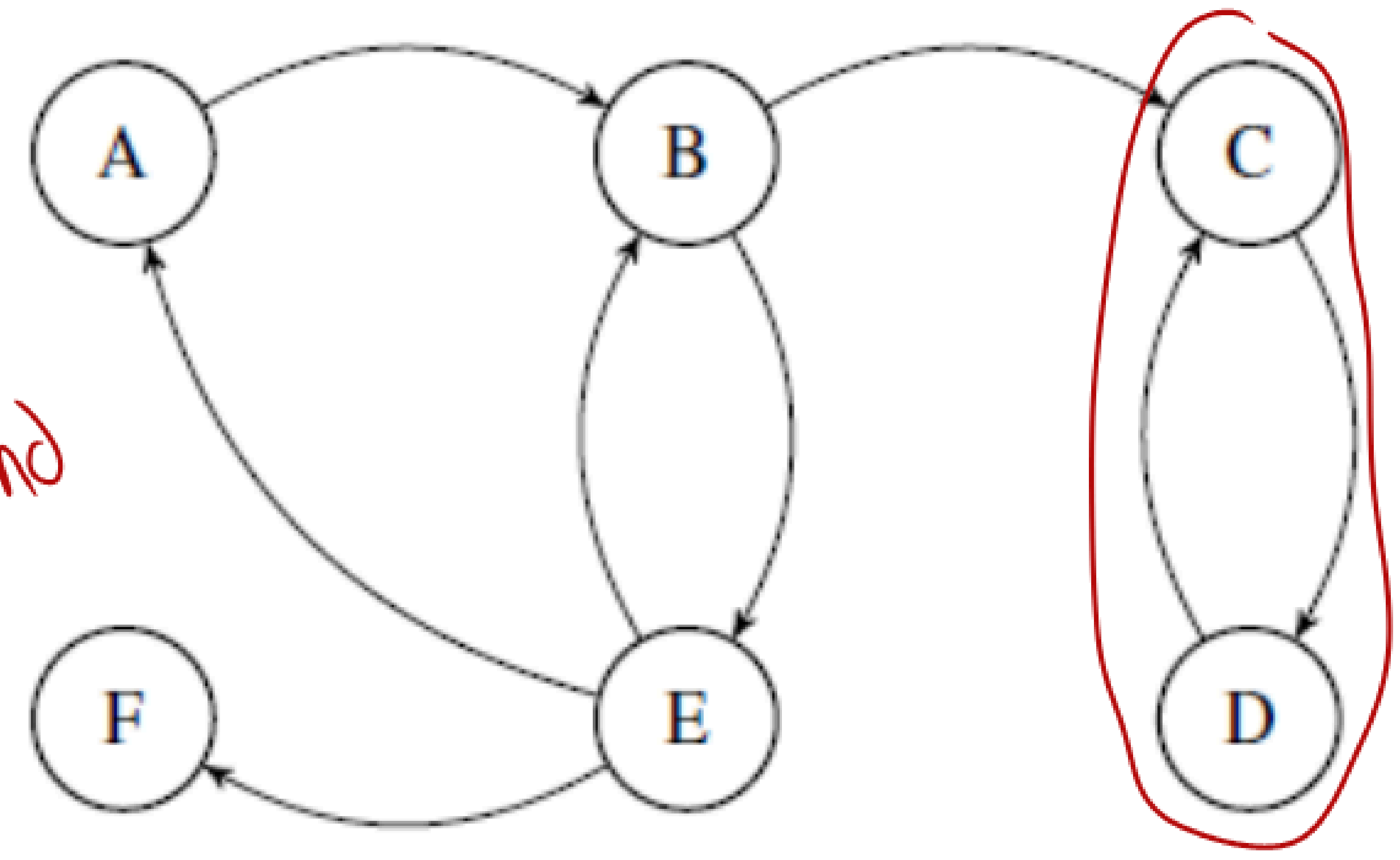
4. Randomly generate a state transition matrix T for a Markov chain with 20 states (identified from 1 to 20) using the Matlab function *rand*.

(a) Confirm that the state transition matrix T is stochastic.

(b) What is the probability that, starting from state 1, the system will be in state 20 after 2 transitions? And after 5? And after 10? And after 100? Display the results in percentage and with 5 decimal places. What do you conclude?

5. Consider a set of web pages, with their interconnecting hyperlinks, given by the following diagram:

dead end



spider trap ↓ ao entrar aqui nunca sai mais

(a) Using the matrix H of hyperlinks, estimate the *pagerank* of each web page at the end of 40 iterations. Recall that you should consider (i) the same transition probability from each page to all next possible pages and (ii) the initial page probability should be the same for all web pages. What is(are) the page(s) with the highest *pagerank* and what is its (their) value(s)?

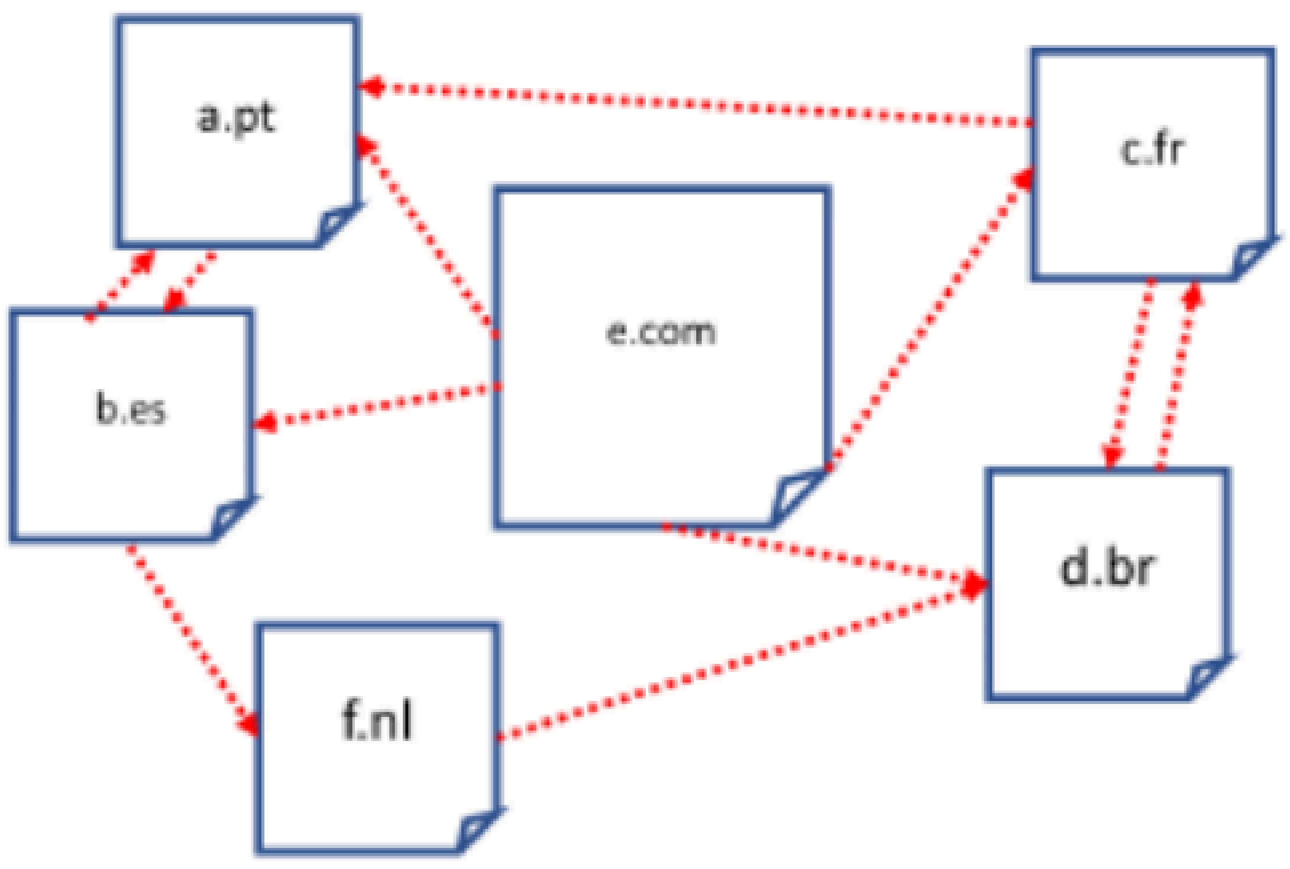
(b) Identify the "spider trap" and the "dead-end" of this set of web pages.

(c) Change matrix H to solve only the “dead-end” problem and determine again the *pagerank* of each web page at the end of 40 iterations.

(d) Solve now also the “spider-trap” problem and determine again the *pagerank* of each web page at the end of 40 iterations (assume $\beta = 0.8$).

(e) Determine now the *pagerank* of each web page considering a minimum number of iterations such that none of the values changes more than 10^{-4} in 2 consecutive iterations. How many iterations are needed? Compare the obtained *pagerank* values with the ones of the previous question. What do you conclude?

6. (Optional) Consider the web pages with the hyperlink connections shown in the figure below:



(a) Use the Google matrix A, with $\beta = 0.85$, to estimate the *pagerank* of each page using an iterative method. Run the method until the difference in *pagerank* between two iterations does not exceed 0.01 in absolute value for all pages. The columns and rows of the matrix must follow the alphabetical order of the page names. Which pages have the lowest and highest *pagerank* and what are their values?

(b) Confirm the estimated *pagerank* values using a non-iterative process.